EX04 - Drag Coefficient - Piecewise Function

Beginner Step-by-Step Guide

This guide follows the real file `ex04_drag_coefficient.py`. It explains the problem first and then the code from the perspective of someone starting Python.

1. What we want to solve

Compute speed of sound and Mach number, then select C_D from three regions.

Problem variables / elements

Name	Meaning
speed	speed [m/s]
T	initial temperature in C
a	speed of sound
M	Mach number
C_D	drag coefficient

Equation or logic on paper

```
T_K = T_C + 273
a = sqrt(gamma * R * T_K)
M = speed / a
C_D = 0.3 if M<0.8; 0.5 if 0.8<=M<1.2; 0.25 otherwise
```

Manual check

With T=20 C, T_K=293 K. a~343 m/s. For speed=500 m/s, M~1.46 and C_D=0.25.

2. What we already know and what is new

Already seen:

- variables
- comparisons
- if

New in this exercise:

- sys.exit()
- C -> K conversion
- np.sqrt()
- if/elif/else as a piecewise function
- format :.2f

3. New Python concepts

1. Stop execution

Syntax: `sys.exit()`

Terminates the script at that point. Here it is used after invalid input is detected.

2. Square root

Syntax: `np.sqrt(value)`

np.sqrt calls a NumPy function and returns the square root.

3. Piecewise function

Syntax: `if / elif / else`

Each branch represents a region. Order matters: elif M<1.2 is reached only after M<0.8 was False.

4. Formatting

Syntax: `{M:.2f}`

:.2f displays two decimal places without changing the stored value.

4. Code step by step

Step 1. Validate inputs

```
if speed < 0:
    print(...)
    sys.exit()

if T <= -273:
    ...
```

They are independent if statements because each failure exits the program.

Step 2. Convert temperature

```
T = T + 273
```

The formula requires absolute temperature. The same name T is reused for Kelvin.

Step 3. Calculate a

```
speed_of_sound = np.sqrt(gamma * R * T)
```

gamma and R are model constants for air.

Step 4. Calculate Mach

```
M = speed / speed_of_sound
```

Mach compares object speed with the speed of sound.

Step 5. Choose C_D

```
if M < 0.8:
    C_D = 0.3
elif M < 1.2:
    C_D = 0.5
else:
    C_D = 0.25
```

The second branch implicitly means 0.8<=M<1.2 because the first branch already failed.

5. Execution flow

```
speed,T
   -> validate
   -> convert T to K
   -> speed of sound
   -> Mach
   -> choose C_D
```

6. Expected result / check

With speed=500 and T=20 C, the file prints Mach~1.46 and C_D=0.25.

7. Common beginner mistakes

- Using Celsius directly in the formula.
- Using three independent if statements for mutually exclusive regions.
- Adding a condition after else.
- Thinking :.2f changes the actual value.

8. If you remember only five things

- Validate before calculating.
- Convert units when the model requires it.
- np.sqrt(...) is a function call.
- if/elif/else maps naturally to piecewise functions.
- Branch order matters.

9. Mini-practice

Change one input. Before running, write a comment predicting what should change and why. Then run the file and compare your prediction with the output.

Completion criterion

Move on when you can explain the problem, equation/logic, new Python concepts and execution flow without simply reading the code.